

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Q3. Scenario: Inventory Management System

Modules:

Add Item, Update Stock, Generate Reports

Question:

Explain the importance of **test cases and test plans** in ensuring system quality. Illustrate with examples from the given system.

Code of Conduct:

I Y. Abhay (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Y. Abhay

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Q3. Inventory Management System

Modules:

- Add Item
- Update Stock
- Generate Reports

Importance of Test Cases and Test Plans

1. Test Plan

A **Test Plan** is a document that describes the overall testing strategy, objectives, scope, resources, schedule, and testing methods for a software system.

Importance of Test Plan

- Defines what needs to be tested.
- Ensures all modules are covered.
- Helps allocate testing resources.
- Reduces testing risks.
- Improves software quality and reliability.

Example (Inventory Management System)

Test Plan Component	Example
Objective	Verify all inventory functions work correctly
Scope	Add Item, Update Stock, Generate Reports
Resources	Testers, Database, Test Environment
Schedule	Unit Testing → Integration Testing → System Testing
Deliverables	Test Cases, Test Reports, Defect Reports

2. Test Case

A **Test Case** is a set of conditions, inputs, and expected results used to verify a specific functionality of the system.

Importance of Test Cases

- Validate system requirements.
- Detect defects early.

- Ensure accurate outputs.
- Improve reliability and user satisfaction.
- Support regression testing after modifications.

Example Test Cases

Module 1: Add Item

Test Case 1

Test Case ID	TC01
Module	Add Item
Input	Item Name = Laptop, Quantity = 10
Expected Result	Item added successfully
Actual Result	Item added successfully
Status	Pass

Test Case 2

Test Case ID	TC02
Module	Add Item
Input	Empty Item Name
Expected Result	Error message displayed
Status	Pass

Module 2: Update Stock

Test Case 3

Test Case ID	TC03
Module	Update Stock
Input	Increase Quantity from 10 to 20
Expected Result	Stock updated successfully
Status	Pass

Test Case 4

Test Case ID	TC04
Module	Update Stock
Input	Negative Quantity (-5)
Expected Result	Validation error
Status	Pass

Module 3: Generate Reports

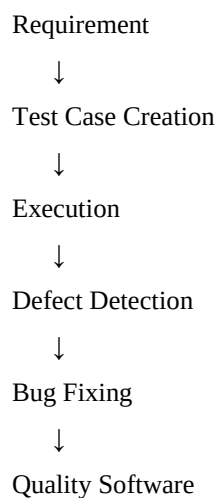
Test Case 5

Test Case ID	TC05
Module	Generate Reports
Input	Monthly Inventory Report
Expected Result	Report generated correctly
Status	Pass

Test Case 6

Test Case ID	TC06
Module	Generate Reports
Input	Empty Database
Expected Result	"No Records Found" message
Status	Pass

How Test Cases Ensure Quality



Benefits

- Ensures correctness of inventory data.
- Prevents stock calculation errors.
- Verifies report accuracy.
- Reduces system failures.
- Improves overall system reliability.

Conclusion

Test Plans provide a structured approach for testing the entire Inventory Management System, while **Test Cases** verify individual functionalities such as Add Item, Update Stock, and Generate Reports. Together, they help identify defects, ensure requirement compliance, improve software quality, and deliver a reliable inventory management solution.